

Math Class 10 Federal Board

Chapter 01 Complex Numbers

For Personal Counselling / Coaching, you can book one to one session with me on <https://www.fakharstemsphere.com/>

Content Domain/ Area	Topics	SLO Number/ Description	Form of Assessment	Cognitive Level (Knowledge, Understanding, Application)	Remarks	Number of Periods required (1 period = 40 minutes)
A: Numbers and Algebra	Complex Numbers	[SLO: M-10-A-01]: Identify complex numbers, complex conjugate, absolute value or modulus of a complex number.	Summative	Knowledge	Question(s) will be asked in the annual examination	15
		[SLO: M-10-A-02]: Apply algebraic properties and perform basic operations on complex numbers.	Summative	Application	Question(s) will be asked in the annual examination	
		[SLO: M-10-A-03]: Demonstrate additive identity and multiplicative identity for the set of complex numbers	Formative	Knowledge	Question(s) will not be asked in the annual examination; however, it will be part of regular teaching practice.	
		[SLO: M-10-A-04]: Find additive inverse and multiplicative inverse of a complex number z.	Summative	Knowledge	Question(s) will be asked in the annual examination	
		[SLO: M-10-A-05]: Demonstrate the following properties of a complex number z.	Summative	Knowledge	Question(s) will be asked in the annual examination	

$ z = -z = \bar{z} = \overline{\bar{z}} $, $\bar{\bar{z}} = z$, $z\bar{z} = z ^2$, $\overline{z_1 + z_2} = \bar{z}_1 + \bar{z}_2$, $\overline{z_1 z_2} = \bar{z}_1 \bar{z}_2$, $\overline{\left(\frac{z_1}{z_2}\right)} = \frac{\bar{z}_1}{\bar{z}_2}$, $z_2 \neq 0$.					
[SLO: M-10-A-06]: Find real and imaginary parts of complex numbers of the type (i) $(x + iy)^n \forall n \in \mathbb{Z}$ (ii) $\left[\frac{x_1 + iy_1}{x_2 + iy_2} \right]^n$ for $(x_2 + iy_2) \neq 0$, Where $n = \pm 1$ and ± 2	Summative	Understanding	Question(s) will be asked in the annual examination		
[SLO: M-10-A-07]: Explain, with examples, how mathematical models and equations are often used to make predictions and test hypotheses in science. [e.g., In physics, mathematical equations are used to describe the motion of objects and the behavior of energy and matter. In chemistry, mathematical models are used to predict the behavior of chemical reactions and the properties of molecules. In biology, mathematical models are used to predict the growth and spread of populations and the spread of disease.]	Summative	Application	Question(s) will be asked in the annual examination		
[SLO: M-10-A-08]: Solve the simultaneous linear equations with complex coefficients.	Summative	Knowledge	Question(s) will be asked in the annual examination		
[SLO: M-10-A-09]: Apply the Geometric interpretation of a complex number.	Summative	Application	Question(s) will be asked in the annual examination		
[SLO: M-10-A-10]: Apply the geometric interpretation of the modulus of a complex number.	Formative	Application	Question(s) will not be asked in the annual examination;		

			however, it will be part of regular teaching practice.	
[SLO: M-10-A-11]: Apply the geometric interpretation of algebraic operations.	Summative	Application	Question(s) will be asked in the annual examination	
[SLO: M-10-A-12]: Solve quadratic equations of the form $pz^2 + qz + r = 0$, $\forall p, q, r \in R$ and $z \in C$ by factorization, quadratic formula, completing square and graphs.	Summative	Knowledge	Question(s) will be asked in the annual examination	

$$x^2 - (1)^2 = 0,$$

$$a^2 - b^2 = (a-b)(a+b)$$

$$(x-1)(x+1) = 0$$

$$x-1=0$$

$$x+1=0$$

$$S.S = \{-1, 1\}$$

$$x = +1$$

$$x = -1$$

$$x^2 - 1 = 0 \Rightarrow x^2 = (1)^2 = 1$$

$$\begin{aligned} (+1)^2 &= 1 \\ (-1)^2 &= 1 \end{aligned}$$

$$x = \pm 1$$

$$S.S = \{-1, 1\}$$

$$x^2 + 1 = 0 \Rightarrow x^2 = -1$$

$$\sqrt{x^2} = \pm \sqrt{-1} = \pm i$$

iota

Q: Rational Nos. $\frac{1}{9}$, $\sqrt{16} = 4$, $\frac{2}{3} = 0.66\ldots$

$$\sqrt{8} = \sqrt{2 \times 4} = 2\sqrt{2}$$

Non-Term.

=

$$\sqrt{8} = \sqrt{2 \times 4} = 2\sqrt{2}$$

Non-Terminating
Repeating
Recurring

- i) Non-Terminating
 - ii) Non-Recurring
- } Irrational N^o

Q'

$$\boxed{\mathbb{R} = \mathbb{Q} \cup \mathbb{Q}'}$$

$\mathbb{Z} \subset \mathbb{Q}$
↓
Set of Integers

$$i = \sqrt{-1} \Rightarrow \boxed{i^2 = -1}$$

$$x^2 + 1 = 0 \Rightarrow x^2 = -1$$

$$x^2 = i^2$$

$$x = \pm i$$

imaginary
solutions

$$S.S = \{-i, i\}$$

MCQS

The real solutions of the

Eqⁿ. $x^2 + 1 = 0$



Ex. $x^2 + 1 = 0$

a) $\{-1, 1\}$ b) 0 c) $\{-i, i\}$ d) $\star$ Non-real

$$x^2 + 4 = 0$$

$$x^2 = -4 = 4(-1) = 4i^2$$

$$x = \pm 2i \rightarrow \text{imaginary no.}$$

$$x^2 + 2 = 0 \Rightarrow x^2 = -2$$

$$x^2 = -1 \times 2 = i^2 \times 2$$

$$x = \pm \sqrt{2} i \rightarrow \text{imaginary}$$

Imaginary Number

An imaginary number is a number that can be written $a+bi$, where a and b are real numbers $b \neq 0$.

$$\frac{a+bi}{\quad} \rightarrow b \neq 0$$

$$\hookrightarrow b \neq 0$$

$a = \text{Real part}$

$b = \text{Imaginary part}$

e.g. $3i$ ✓, $2+3i \rightarrow$ imaginary number

If $a=0$, (bi)

$\hookrightarrow$ Pure Imaginary N°.

Examples

$3i, -2i, 5i, i, -i, \frac{3}{2}i$

Pure Imaginary N°s

Complex Numbers

When a and b are real numbers and b is allowed to be 0, the number $a+bi$ is said to be complex.

A complex number is any number that can be written in the form of $a+bi$, where a and b are any real numbers and both a and b can be zero.

$$\boxed{a+bi}$$

a " " zero
 b can be zero

$$\boxed{a+bi} \rightarrow \text{Complex No.}$$

Example $z = \textcircled{7} = 7 + 0i$ ^{Real No.}

$$z = a + bi$$

$$a = 7, \quad b = 0$$

$$\mathbb{R} \subset \mathbb{C}$$

* Set of Real #s are Subset of Complex Nos.

* Every Real No. is Complex No. but with imaginary part $\rightarrow$

Examp, $2, -5, -7, \frac{5}{2}, \sqrt{5}$
are all Complex No.

' ' are all Compiler no

Example

$$\sqrt{-x}$$

$$\sqrt{-1 \times x} = \sqrt{-1} \times \sqrt{x}$$

$$\sqrt{-x} = \sqrt{x} i \quad \rightarrow \text{Calculator}$$

$$i) \sqrt{-5} = \sqrt{5} i$$

MCA

d. $-\sqrt{-64}$

e. $3i \times 5i$

f. $\sqrt{-5} \times \sqrt{-20}$

$$\begin{aligned} &\downarrow \\ &= -\sqrt{64 \times -1} \\ &= -\sqrt{64} \times \sqrt{-1} \\ &= -8i \end{aligned}$$

$$\begin{aligned} &\downarrow \\ &= 15i^2 \\ &= 15(-1) \\ &= -15 \end{aligned}$$

$$\begin{aligned} &\downarrow \\ &= \sqrt{5} \times i \times \sqrt{20} \times i \\ &= \sqrt{100} i^2 \\ &= -10 \end{aligned}$$

$$Z = a + bi = (a, b)$$

Addition of Complex Nrs.

Real part

imaginary part

$$Z_1 = a + bi, \quad Z_2 = c + di$$

$$Z_1 + Z_2 = (a + c) + \underline{bi + di}$$

$$Z_1 + Z_2 = \underbrace{(a + c)}_{\text{Real part}} + \underbrace{(b + d)i}_{\text{Imaginary part}}$$

$$Z_1 = (a, b), \quad Z_2 = (c, d)$$

$$Z_1 + Z_2 = (a + c, \underline{b + d})$$

$$Z_1 + Z_2 = \underline{a + c + (b + d)i}$$

Real part of
 $Z = Z_1 + Z_2 = a + c$

$$z = z_1 + z_2$$

Real($z_1 + z_2$) or

$$\text{Re}(z_1 + z_2) = a + c$$

$$\text{Im}(z_1 + z_2) = b + d$$

Difference of Complex nos.

$$z_1 = a + bi, \quad z_2 = c + di$$

$$z_1 - z_2 = a - c + (b - d)i$$

$$3x + 51 = 84$$

Example Simplify

a. $(7 + 5i) + (4 + 3i)$

$$= 11 + 8i$$

b. $(5 + 6i) - (7 - 4i)$

$$\begin{aligned} &= 5 + 6i - 7 + 4i \\ &= -2 + 10i \end{aligned}$$

Check point

$$\sqrt{-7} \times \sqrt{-5} = ?$$

$$\sqrt{-2} \times \sqrt{-5} = ?$$

a) $\sqrt{10}$ b) $\sqrt{-10}$ c) $\sqrt{10}i$ d) $-\sqrt{10}$

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

$$\sqrt{-2} \times \sqrt{-5} = \sqrt{(-2) \times (-5)} = \sqrt{10}$$

$$i \times i = i^2 = -1$$

Example: Multiply and simplify:

a. $\sqrt{-2}\sqrt{-11}$

b. $-2i \cdot 5i$

c. $-3i(2 - 8i)$

d. $\sqrt{-9}\sqrt{-36}$

e. $(3 - 2i)(1 + 5i)$

f. $(5 - i)(9 - 3i)$

$$\sqrt{-2} \times \sqrt{-11} = \sqrt{22}$$

$$\begin{aligned} a) &= \sqrt{2}i \times \sqrt{11}i \\ &= \sqrt{22}i^2 \\ &= -\sqrt{22} \end{aligned}$$

$$\begin{aligned} b) &= 10 \\ &= 10i^2 \\ &= -10 \end{aligned}$$

$$\begin{aligned}
 a) &= \sqrt{2} i \times \sqrt{11} i \quad (= 10) \\
 &= \sqrt{22} i^2 \\
 &= -\sqrt{22}
 \end{aligned}
 \quad \left| \begin{aligned}
 &= 10i^2 \\
 &= \textcircled{+10}
 \end{aligned} \right.$$

c. $-3i(2 - 8i)$

f. $(5 - i)(9 - 3i)$

$$a(b - c)$$

$$\begin{aligned}
 c) & -3i(2 - 8i) \\
 &= (-3i)(2) - (-3i)(8i) \\
 &= -6i + 24i^2
 \end{aligned}$$

$$\begin{aligned}
 &= -6i - 24 \\
 &= \boxed{-24 - 6i}
 \end{aligned}$$

$$\begin{aligned}
 i &= \sqrt{-1} \\
 i^2 &= -1
 \end{aligned}$$

f. $(5 - i)(9 - 3i)$

$$\underline{(a + bi)(c + di)}$$

$$\begin{aligned}
 &= 5(9 - 3i) - i(9 - 3i) \\
 &= 45 - 15i - 9i + 3i^2
 \end{aligned}$$

$$= 45 - \underline{\underline{15i}} - 9i + 5i$$

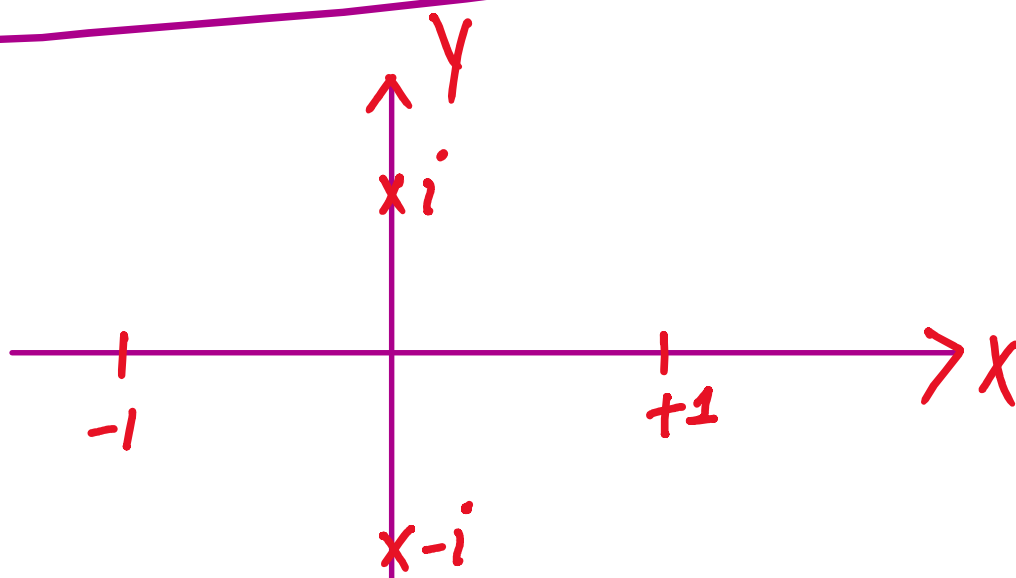
$$= 45 - 3 - 15i - 9i \quad (\because i^2 = -1)$$

$$= 42 - 24i$$

Thus,

$$(5-i)(9-3i) = 42 - 24i$$

$x+yi$
(x, y)



$$1 \rightarrow 1(i) = i$$

$$i \xrightarrow{90^\circ} i^2 = (-1)$$

$$-i \cdot i = -i^2 = -(-1) = 1$$

$$\boxed{1, i, -1, -i}$$

$$i, -1, -i, 1$$

$i \times (No) = \text{Anticlockwise Rotation by } 90^\circ$

$$\begin{aligned} i^1 &= i \\ i^2 &= -1 \end{aligned}$$

$$i = \sqrt{-1} \Rightarrow i^2 = -1 \quad i^{1+2}$$

$$i^3 = i \cdot i^2 = -i$$

$$i^4 = (i^2)^2 = (-1)^2 = +1$$

$$i^5 = i \cdot i^4 = i \cdot 1 = i$$

MCQs

$$\rightarrow i^2 = -1 \quad i^{10} = i^{2 \times 5} = (i^2)^5$$

Example: Simplify:

a. i^{11}

b. i^{22}

c. i^{61}

d. i^{100}

$$a) i^{11} = i^{10+1} = i \cdot i^{10} = i (i^2)^5 = i (-1)^5 = -i$$

$$1, \dots, 22 \quad 1, 2, \dots, 11 \quad 1, \dots, 11 \quad 1, \dots, 11$$

$$h) i^{22} = (i^2)^{11} = (-1)^{11} = -1$$

$$c) i^{61} = i \cdot i^{60} = i \cdot (i^2)^{30} = (-1)^{30} \cdot i = 1 \cdot i = i$$

MCP

$$i^{-11} ?$$

$$\frac{1}{i^{19}} = ?$$

$$a) i^{-11} = \frac{1}{i^{11}} = \frac{1}{i^{10} \cdot i} = \frac{1}{(i^2)^5 \cdot i} = \frac{1}{(-1)^5 \cdot i}$$

$$i^{-11} = \frac{1}{-i} = -\frac{1}{i} = \frac{-1}{i}$$

$$\Rightarrow i^{-11} = 1$$

$$= i$$

$$\text{OR } i^{-11} = \left\{ -\frac{1}{i} \right\} = \frac{-1}{i} \times \frac{i}{i} = \frac{-i}{i^2}$$

$$= \frac{-i}{-1} = i$$

$$\frac{1}{i^{19}} = \frac{1}{i \cdot (i^2)^9} = \frac{1}{i \cdot (-1)^9} = \frac{\boxed{-1}}{i} = \frac{i^2}{i} = \boxed{i}$$

M(QS)
2

$$i^{-99} = \frac{1}{i^{99}} = \frac{1}{i^{98} \cdot i}$$

$$= \frac{1}{(i^2)^{49} \cdot i} = \frac{-1}{i} = \boxed{i}$$

Example: Write the expressions in the form of $a+ib$.

a. $2+i^3$

b. $19+7i^2$

c. $i^{12}+i^{15}$

d. $4i^3+8i^6$

$$\begin{aligned} &= 2 + \boxed{i^2} \cdot i \\ &= 2 - i \end{aligned}$$

$$\boxed{a=2, b=-1}$$

$$\begin{matrix} a & + & bi \\ \uparrow & & \uparrow \\ 2 & - & i \end{matrix}$$

$$\begin{aligned} &= 19 + 7(-1) \\ &= 19 - 7 \\ &= \boxed{12} \end{aligned}$$

$$a=12, b=0$$

$$\begin{aligned} &= (i^2)^6 + i \cdot (i^2)^7 \\ &= (-1)^6 + (-1)^7 \cdot i \\ &= \boxed{1 - i} \end{aligned}$$

2-6

d. $4i^3 + 8i^6$

$$= 4(-1)i + 8 \cdot (i^2)^3 \quad (-1)^3$$

$$= -4i - 8$$

$$= -8 - 4i$$

$$\underbrace{a = -8}_{\text{Re}}, \underbrace{b = -4}_{\text{Im}}$$

Conjugate of Complex No.

$$Z = a + bi = (a, b)$$

$$\bar{Z} = a - bi = (a, -b)$$

↓
Conjugate of Z

Example: Find the conjugate:

a. $5+9i$

$\downarrow$
 $5-9i$

b. $13-5i$

$13+5i$

c. $-5-7i$

$\downarrow$
 $-5+7i$

d. $11i$

$\downarrow$
 $-11i$

Let $z = 2$

$\boxed{\bar{z} = 2}$

MCQ

$\overline{-2-5i} = ?$

a) $-2-5i$, b) $2+5i$ c) $-2+5i$
d) $2-5i$

Division of Complex Numbers

$$\frac{z_1}{z_2} = \frac{x_1 + iy_1}{x_2 + iy_2} \times \frac{x_2 - iy_2}{x_2 - iy_2}$$

Example: Divide and simplify to the form $a+ib$.

a. $\frac{-7+3i}{2-5i}$

b. $\frac{11+5i}{10i}$

c. $\frac{1-7i}{3+4i}$

Sol: Let $z = \frac{-7+3i}{2-5i}$

$$= \frac{-7+3i}{2-5i} \times \frac{2+5i}{2+5i}$$

$$= \frac{-7(2+5i) + 3i(2+5i)}{(2)^2 - (5i)^2}$$

$$= \frac{-14 - 35i + 6i + 15i^2}{4 - 25i^2}$$

$$= \frac{-14 - 15 - 29i}{4 - 25(-1)} \quad (\because i^2 = -1)$$

$$= \frac{-29 - 29i}{29}$$

$$= \frac{-28}{28} - \frac{28i}{28}$$

$$\boxed{Z = -1 - i}$$

$$\operatorname{Re}(z) = -1, \operatorname{Im}(z) = -1$$

Hence $\boxed{\frac{-7+3i}{2.5i} = -1 - i}$ ~~Ans~~

$$b. \frac{11+5i}{10i} \times \frac{+10i}{+10i}$$

$$= \frac{110i + 50i^2}{100i^2}$$

$$= \frac{-50 + 110i}{-100}$$

$10i = 0 + 10i$
 $\therefore$ Therefore
 Because
 $(\because i^2 = -1)$

...

$$= \frac{-50}{-100} + \frac{110i}{-100}$$

$$= \boxed{\frac{1}{2} - \frac{11}{10}i}$$

Thus,

$$\frac{11+52i}{10i} = \frac{1}{2} - \frac{11}{10}i$$

$$i^2 = -1$$

ii. $6\sqrt{-4}$

iii. $\sqrt{-\frac{4}{9}}$

v. $4 - \sqrt{-60}$

vi. $\sqrt{-8}\sqrt{-2}$

ii)

$$= 6 \times \sqrt{4i^2}$$

$$= 6 \times 2i$$

$$= 12i$$

iii)

$$= \sqrt{\frac{4}{9} \times i^2}$$

$$= \sqrt{\frac{(2)^2 \cdot i^2}{(3)^2}}$$

$$= \frac{2}{3}i$$

15x4

$$v) 4 - \sqrt{-60}$$

$$\sqrt{15 \times 4}$$

$$= 4 - \sqrt{15 \times 4 i^2}$$

$$= 4 - 22\sqrt{15}$$

$$\boxed{4 - 2\sqrt{15} i}$$

$$\boxed{-1 = i^2}$$

$$vi) \sqrt{8} \times \sqrt{-2}$$

$$= \sqrt{8 i^2} \times \sqrt{2 i^2}$$

$$= \sqrt{8} i \times \sqrt{2} i$$

$$= \sqrt{16} \overset{i^2}{\downarrow} \overset{-4}{\bullet}$$

$$\sqrt{(a+b)^2} = a^2 + 2ab + b^2$$

$$viii. (3+2i)^2$$

viii. $(3+2i)^2$

$$= (3)^2 + 2(3)(2i) + (2i)^2$$

$$= 9 + 12i + 4i^2$$

$$= 9 + 12i - 4$$

$$= \underline{5 + 12i}$$

$$(a+b)^2 = a^2 + 2ab + b^2$$

$$(3+2i)(3+2i)$$

$$(\because i^2 = -1)$$

$$\underline{(a+b)^2 = (-a-b)^2}$$

x. $\underline{(-5-3i)^2}$

xi. $(1+\sqrt{2}i)(1-\sqrt{3}i)$

xii. $(\sqrt{2}+i)(\sqrt{2}-i)$

$$= \underline{-(5+3i)}^2$$

$$= (5+3i)^2$$

$$= (5+3i)(5+3i)$$

$$= 25 + 15i + 15i + 9i^2$$

$$= \underline{25 - 9 + 30i} \quad \Delta$$

$$= 25 - 17 = 8$$

$$= \sqrt{16 + 30i}$$

→

$$\sqrt{a} \times \sqrt{b} = \sqrt{ab}$$

xi. $(1 + \sqrt{2}i)(1 - \sqrt{3}i)$

$$= 1(1 - \sqrt{3}i) + \sqrt{2}i(1 - \sqrt{3}i)$$

$$= 1 - \sqrt{3}i + \sqrt{2}i - \sqrt{2}\sqrt{3}i^2$$

$$= \boxed{\underbrace{1 + \sqrt{6}}_{\text{Real}} + \underbrace{(\sqrt{2} - \sqrt{3})}_{\text{Imag.}}i}$$

xii. $(\sqrt{2} + i)(\sqrt{2} - i)$

$$= (\sqrt{2})^2 - i^2$$

$$= \underline{2 - (-1) = 2 + 1 = 3}$$

3. Simplify:

i. $i^9 = i$

ii. $i^{13} = i$

iii. i^{28}

iv. $(-i)^{21}$

v. $(3i)^3$

vi. $(-2i)^4$

i) $i^9 = i^8 \cdot i = (i^2)^4 \cdot i = (-1)^4 \cdot i = i$

ii) $i^{13} = i^{12} \cdot i = (i^2)^6 \cdot i = i$ ✓

iii) $i^{28} = (i^2)^{14} = (-1)^{14} = 1$ ✓

iv) $(-i)^{21} = (-1)^{21} \cdot i^{21}$
 $= -i^{20} \cdot i = -(i^2)^{10} \cdot i = -(-1)^{10} \cdot i = -i$

v) $(3i)^3 = 27i^3 = 27(i^2)i = -27i$

vi) $(-2i)^4 = (-2)^4 \cdot i^4$
 $= 16(i^2)^2 = 16(-1)^2 = 16$

v. $i^5 + i^7$

$$\begin{aligned}
 &= i^4 \cdot i + i^6 \cdot i \\
 &= (i^2)^2 \cdot i + (i^2)^3 \cdot i \\
 &= (-1) \cdot i + (-1) \cdot i \\
 &= i - i = 0
 \end{aligned}$$

vi. $i^{74} - i^{100}$

$$\begin{aligned}
 &= (i^2)^{37} - (i^2)^{50} \\
 &= (-1)^{37} - (-1)^{50} \\
 &= -1 - 1 = -2
 \end{aligned}$$

iii. $\frac{3-i\sqrt{5}}{3+i\sqrt{5}}$

$$\begin{aligned}
 &\frac{-x \cdot x}{(3)(i\sqrt{5})} \\
 &= \frac{-(\sqrt{5})^2}{(3)(i\sqrt{5})}
 \end{aligned}$$

$$= \frac{9 - 3\sqrt{5}i - 3\sqrt{5}i + 5i^2}{(3)^2 - (i\sqrt{5})^2}$$

$$= \frac{9 - 5 - 2(3\sqrt{5})i}{9 - 5i^2}$$

$$= \frac{4 - 6\sqrt{5}i}{4}$$

$$\begin{aligned}
 & \overline{9+5} \\
 = & \frac{4 - 6\sqrt{5}i}{14} \\
 = & \frac{4}{14} - \frac{6\sqrt{5}i}{14} \\
 = & \boxed{\frac{2}{7} - \frac{3\sqrt{5}i}{7}}
 \end{aligned}$$

viii. $\frac{a+ib}{a-ib}$

$$= \frac{a+ib}{a-ib} \times \frac{a+ib}{a+ib}$$

$$= \frac{(a+ib)^2}{(a-ib)(a+ib)}$$

$$a^2 + (ib)^2 + 2(a)(ib)$$

$$= \frac{a^2 + (ib)^2 + 2(a)(ib)}{a^2 - (ib)^2}$$

$$= \frac{a^2 - b^2 + 2abi}{a^2 - b^2 i^2}$$

$$= \frac{a^2 - b^2 + 2abi}{a^2 + b^2}$$

$$\frac{a+ib}{a-ib} = \frac{a^2-b^2}{a^2+b^2} + \frac{2ab}{a^2+b^2}i$$

$$\frac{a-ib}{a+ib} = \frac{a^2-b^2}{a^2+b^2} - \frac{2ab}{a^2+b^2}i$$

EXTRA Test

$$\frac{3 - \sqrt{5}i}{3 + \sqrt{5}i} = \frac{9 - 5}{9 + 5} - \frac{2(3)(\sqrt{5})i}{147}$$

$$= \frac{4}{14}$$

$$= \frac{2}{7}$$

ix. $\frac{1+i}{(1-i)^2}$ x. $\frac{(2+2i)^2}{(1+i)^2}$

ix) $\frac{1+i}{(1-i)^2}$

$$(a-b)^2 = a^2 + b^2 - 2ab$$

$$= \frac{1+i}{1+i^2-2(1)(i)} = \frac{1+i}{-2i}$$

$$= \frac{1+i}{-2i} \times \frac{2i}{2i}$$

$$= \frac{2i + 2i^2}{-2i^2}$$

$$= \frac{2i + 2i^{-}}{-4i^2}$$

$$= \frac{-2 + 2i}{4} = -\frac{2}{4} + \frac{2i}{4}$$

$$= -\frac{1}{2} + \frac{1}{2}i$$

x. $\frac{(2+2i)^2}{(1+i)^2}$

~~$\frac{x^2}{y^2} = \frac{x}{y}$~~
Wrong.

$$= \left(\frac{2+2i}{1+i} \right)^2$$

$$= \left(\frac{2(1+i)}{(1+i)} \right)^2 = 2^2 = \textcircled{4}$$

Exercise 1.1 End